MODULE 10 · DOMAIN 2 - CLAUDE MODELS, PROMPTING AND CONTEXT ENGINEERING

Prompt Reuse: Caching, Modular Prompts, Skills

How prompt caching really works and what breaks it, shared prompt modules, and when to package a procedure as a Skill.

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Write it once, and make every request **reuse it**: cached for cost, modular for change, packaged for teams.

How prompt caching works

- **Mark a stable prefix.** Later requests with an identical prefix read it from cache.
- **The output is unchanged.** Only cost and time to first token improve.
- **Up to four breakpoints,** or automatic caching with one top-level setting.
- **Below 512 tokens on current models,** nothing is cached, and no error says so.
- **Check the usage fields:** cache creation and cache read tokens.

> ON THE EXAM
>
> Caching changes the bill and the latency, never the answer.

What caching costs, October 2026

ITEM	MULTIPLE OF BASE INPUT PRICE
Cache write, 5-minute lifetime	1.25x
Cache write, 1-hour lifetime	2x
Cache read, most models	0.1x
Cache read, Opus 5.5 and Sonnet 5.5	0.05x
Cache read, Fable 5.1	0.025x

ON THE EXAM

The 5-minute lifetime refreshes free on every hit. Choose 1 hour when gaps between requests are longer.

Order decides what can be cached

01
Tools
Definitions. Change one, lose everything.

02
System
Instructions, policy, examples.

03
Earlier messages
The conversation so far.

04
Breakpoint
Everything above is cached.

05
New input
The new turn, never cached.

ON THE EXAM

A change invalidates its own level and everything after it. Stable first, volatile last.

What silently breaks the cache

CHANGE	WHAT IT INVALIDATES
Any edit to tool definitions	Everything
A timestamp or user name in the system prompt	The system prompt and all after
Turning web search or citations on or off	System prompt and messages
Changing effort or thinking settings	Messages
Volatile content placed before stable content	Everything after it

ON THE EXAM

A cache hit needs an identical prefix. One changing token near the top costs you all of it.

Cache, retrieve or summarise?

SITUATION	BEST REUSE
A large, stable document queried many times in a session	Cache it whole
The same instructions on thousands of requests	Cache the system prompt
A corpus too big for the window, or changing often	Retrieve relevant passages
Only the gist is needed later	Summarise it

ON THE EXAM

Caching keeps everything and costs a fraction to reread. Summaries and retrieval drop content.

Caching alongside everything else

- **Stacks with Batch:** cache hits in batches are best effort; the 1-hour lifetime helps.
- **Cached reads do not count** towards input-token rate limits.
- **Time to first token improves** for long prompts.
- **Isolation:** never shared across organisations; per workspace on the Claude API.

> ON THE EXAM
>
> Caching is a cost lever, a latency lever and a rate-limit lever at once.

Modular prompts

- **Shared modules:** house style, safety rules, output schemas, maintained once.
- **Assembled into templates,** with stable modules first so they stay cacheable.
- **Versioned, with one owner each,** and tested when they change.
- **Copy-and-paste prompts drift** apart within months.

> ON THE EXAM
>
> Change a shared rule once and every prompt that uses it changes with it.

Skills: progressive disclosure

Level 1: metadata

Name and description
Always loaded, small

Level 2: instructions

The SKILL.md body
Loaded when relevant

A Skill

SKILL.md instructions
Optional scripts and files

A folder Claude loads when relevant

Level 3: files

Scripts, references
Opened as needed

THE TAKEAWAY

Until a Skill is needed, only its short description costs tokens.

Where should it live?

PUT IT IN	WHEN IT IS
System prompt	A rule every request needs
Skill	A procedure, used when relevant, shared across workflows
MCP server or tool	Access to a live system or data
Retrieval corpus	Large reference material, looked up per question

ON THE EXAM

Anthropic: if you are explaining how to do something, that is a Skill. If Claude needs to access something, that is MCP.

Governing shared Skills

- **One owner, pinned versions,** and a few test queries per Skill.
- **Separation of duties:** authors do not approve their own Skills.
- **Install only from trusted sources;** scripts run with real access.
- **Custom Skills do not sync across surfaces:** publish to each one.
- **Currency check:** Skills are not ZDR-eligible; on the API they need code execution.

> ON THE EXAM
>
> A Skill is shared code and shared instructions. Govern it like both.

Mereside's agent assist builds each prompt as: system instructions, then the live call transcript, then the 70-page lending policy of about 45,000 tokens, with a cache breakpoint at the end.

Caching is switched on, but the usage fields show zero cache reads on every request, and costs are high.

What is the most effective fix?

A Report a fault to Anthropic, because caching is not working

B Switch to the one-hour cache lifetime

C Move the policy above the transcript, with the breakpoint after the policy

D Summarise the policy so it costs less to send

◆ PAUSE AND DECIDE

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What is the most effective fix?

A Report a fault to Anthropic, because caching is not working

Caching works as designed. The prefix is never identical.

B Switch to the one-hour cache lifetime

A longer lifetime cannot help a prefix that changes every time.

C Move the policy above the transcript, with the breakpoint after the policy

Stable content before volatile: the policy becomes a reusable prefix.

D Summarise the policy so it costs less to send

Loses exact policy wording, and still caches nothing.

THE SIGNAL

Volatile content before stable content: zero cache reads. Reorder, stable first.

> Tallyhawk's support system prompt begins with the line "Current time:" followed by a timestamp to the second, then about 20,000 tokens of stable instructions and examples.
>
> A breakpoint is set after the instructions, but the cache hit rate is close to zero.

What should Corvane change?

A Move the timestamp out of the system prompt and into the latest user turn

B Add three more cache breakpoints inside the instructions

C Switch to the one-hour cache lifetime

D Remove the examples to shorten the prompt

◆ PAUSE AND DECIDE

What should Corvane change?

A Move the timestamp out of the system prompt and into the latest user turn

The changing token leaves the prefix; 20,000 stable tokens become cacheable.

B Add three more cache breakpoints inside the instructions

Every breakpoint still sits after the changing timestamp.

C Switch to the one-hour cache lifetime

The prefix differs every second. Lifetime is irrelevant.

D Remove the examples to shorten the prompt

Cuts quality and still caches nothing.

THE SIGNAL

One changing value at the top of the prompt: move it below the breakpoint.

> A lawyer at a commercial firm reviews one 120-page contract, about 50,000 tokens, in an afternoon. She asks 20 to 30 questions, in bursts separated by gaps of 10 to 20 minutes.
>
> Answers must consider the whole contract, and the firm wants to cut cost per question.

Which approach fits best?

A. Summarise the contract once, and answer every question from the summary

B. Index the contract in a vector store and retrieve passages per question

C. Cache the contract as a prefix with the one-hour lifetime

D. Cache the contract with the default five-minute lifetime

◆ PAUSE AND DECIDE

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Which approach fits best?

A. Summarise the contract once, and answer every question from the summary

The whole contract must be considered. A summary drops clauses.

B. Index the contract in a vector store and retrieve passages per question

Answers see fragments, not the whole contract.

C. Cache the contract as a prefix with the one-hour lifetime

Whole contract every time, and the cache survives the gaps.

D. Cache the contract with the default five-minute lifetime

Expires during the 10 to 20 minute gaps, so it is rewritten again and again.

THE SIGNAL

A stable document, many questions, gaps longer than five minutes: cache it, for an hour.

Mereside's complaint-response procedure has fixed steps, mandatory regulatory wording and approved examples. Three workflows need it: agent assist, the complaints drafting pipeline, and the quality-review tool.

The procedure changes about five times a year, and each workflow currently holds its own copy.

How should the procedure be packaged?

A Keep a copy in each workflow's system prompt, updated by each team

B One Skill owned by the complaints compliance team, versioned, and used by all three workflows

C An MCP server that returns the procedure text on request

D Add the procedure to the retrieval corpus of policy documents

◆ PAUSE AND DECIDE

How should the procedure be packaged?

A. Keep a copy in each workflow's system prompt, updated by each team

Three copies, five changes a year. guaranteed drift.

B. One Skill owned by the complaints compliance team, versioned, and used by all three workflows

One governed procedure, changed once, used everywhere.

C. An MCP server that returns the procedure text on request

MCP is for access to live systems, not for how-to knowledge.

D. Add the procedure to the retrieval corpus of policy documents

Retrieval may return fragments of a procedure that must be followed whole.

THE SIGNAL

A how-to procedure, shared across workflows, changing often: one owned Skill.

> Tallyhawk wants its support agent to check a customer's current subscription status in the billing service while answering billing questions.
>
> A developer proposes writing a Skill that describes how billing works.

What should Corvane recommend?

A The Skill as proposed, with detailed billing rules

B Paste the billing documentation into the system prompt

C Add the billing documentation to the retrieval corpus

D A read-only tool or MCP server for the billing service, plus a Skill only if procedure guidance is also needed

◆ PAUSE AND DECIDE

What should Corvane recommend?

A. The Skill as proposed, with detailed billing rules
Explains billing, but cannot see this customer's live status.

B. Paste the billing documentation into the system prompt
Documentation is not live data.

C. Add the billing documentation to the retrieval corpus
Still documentation, still not live data.

D. A read-only tool or MCP server for the billing service, plus a Skill only if procedure guidance is also needed
Live access is a tool job; know-how is a Skill job.

THE SIGNAL

Needs current, per-customer data: that is access, so a tool or MCP. A Skill cannot fetch it.

> Tallyhawk's support agent sends about 30 tool definitions and a 15,000-token system prompt on every request. Its deployment pipeline regenerates the tool list on each release, in a different order each time.
>
> Corvane wants to cut input cost without changing answer quality.

Which two changes will cut input cost without changing quality? Choose two.

- A Emit tool definitions in a fixed, deterministic order on every release
- B Place a cache breakpoint after the system prompt
- C Start the system prompt with the customer's name, for a personal touch
- D Remove the examples from the system prompt
- E Move every conversation to the lightest tier

◆ PAUSE AND DECIDE

Which two changes will cut input cost without changing quality? Choose two.

A Emit tool definitions in a fixed, deterministic order on every release

Tools sit first in the prefix; a stable order keeps the cache alive.

B Place a cache breakpoint after the system prompt

Tools plus 15,000 tokens become a cheap cached read.

C Start the system prompt with the customer's name, for a personal touch

A per-customer value at the top breaks the cache for everyone.

D Remove the examples from the system prompt

Cuts tokens and changes quality.

E Move every conversation to the lightest tier

Cuts cost and risks quality on hard cases.

THE SIGNAL

Cost cuts with no quality change: a stable prefix, and a breakpoint after it.

In five lines

- **Caching:** identical prefix, cheaper and faster, same answer. Check the usage fields.
- **Order is everything:** tools, system, messages. Stable first, volatile last.
- **Timestamps and names at the top** are the classic silent cache-killers.
- **Cache** a stable document queried often; **retrieve** a big or changing corpus.
- **Skills hold procedures; MCP gives access.** One owned, versioned Skill beats many copies.

COMING UP

Domain 2 complete. Next, Domain 3: capability bloat and least privilege.